

Arrival choices

Previous lesson reminder:

Start with 1 and 2. Try 3 and 4 when ready. Reading and exact scribing are available.

1 What hazard followed seafloor displacement?

Support: Use the reminder strip or talk it through with an adult.

My response

2 Which meaning fits climate?

Support: Choose: Weather patterns over a long period OR A process or event with potential to cause harm.

My response

3 What happens when fossil fuels are burned?

Support: Use the model evidence anchor B or read one relevant card together.

My response

4 How do emissions reduction and shade differ?

Support: Give a reason when ready.

My response

Stretch: use a precise example or explain which detail supports your answer.

Evidence cards

Read, point or listen. Keep the source label with any detail you use.

A Zones and events

Broad temperature zones relate to latitude. Near the equator is generally warmer; high latitudes are generally colder. Local climate also depends on altitude and other factors. Earthquakes relate mainly to plate boundaries, not climate zones.

Source: Authored map explanation

Limit: Broad bands simplify climate; they are not a detailed climate classification.

B Human influence

Burning fossil fuels adds carbon dioxide. Greenhouse gases slow heat loss to space. Human activity is the main cause of recent global warming.

Source: NASA Causes of climate change

Limit: One local hot or cold day does not establish the global trend.

C Two responses

Reducing fossil-fuel use tackles a cause of warming. Shade, water access and cooler buildings can reduce harm during heat. These responses do different jobs.

Source: Authored examples based on climate mechanisms

Limit: Their suitability depends on the place and the needs of people there.

D Community accounts

Fictional account 1: a faith group explains caring for creation as its reason for planting shade trees.

Fictional account 2: a non-religious residents' group gives human wellbeing as its reason for a cool-room project.

Source: Two authored teaching accounts with explicitly stated values

Limit: These are not reports of real organisations and do not represent every religious or non-religious person.

Visual resource

Use this visual with the evidence cards and enquiry questions.



Source: Natural Earth public-domain outlines, labels simplified. Terms of use:
<https://www.naturalearthdata.com/about/terms-of-use/> (recorded in Sources and checks).

A detail I noticed and what it helps me explain

Data for the enquiry

Selected global annual temperature anomalies

Year	Anomaly °C
1980	0.26
1990	0.45
2000	0.39
2010	0.72
2020	1.01
2024	1.29

Units: °C relative to 1951–1980

Source: NASA GISTEMP v4 annual J-D column; selected values checked 19 September 2026. The full series can be revised.

Limit: Selected years simplify the record. Use the full annual series to examine variability. An anomaly is a difference from the baseline, not the absolute temperature.

My labelled graph or data notes

Shared investigation

1. Read the dated temperature data and its baseline.
2. Distinguish the long-term pattern from single-year variation.

Work with a partner or adult, or use your own table. One person suggests; the other checks the evidence. Swap after one turn. Passing is allowed.

Use the evidence cards and any visual sheet. Follow the two shared steps above, then record one decision.

Our explanation

Support: Read one evidence card together. Highlight a useful detail. Rehearse with: From ____ to ____, the data _____. Burning ____ adds _____. One response is _____.

Further challenge: Explain why a short data selection cannot establish every local effect.

Supported independent task

Complete this route only. Investigate causes and effects of climate change.

1. Describe a pattern in the supplied climate data.
2. Explain a cause and effect using the source.
3. Compare one response that tackles a cause with one that reduces harm.

Support: Read one evidence card together. Highlight a useful detail. Rehearse with: From ____ to ____, the data _____. Burning ____ adds _____. One response is _____. Complete the organiser one space at a time. From ____ to ____, the data _____. Burning ____ adds _____. One response is _____.

Data source and date or capture time

Variables units and graph labels

A pattern supported by values

What the data cannot establish

An adult may read or scribe your exact response. They should not choose the answer.

Standard independent task

Complete this route only. Investigate causes and effects of climate change.

1. Describe a pattern in the supplied climate data.
2. Explain a cause and effect using the source.
3. Compare one response that tackles a cause with one that reduces harm.

Support: From ____ to ____, the data _____. Burning ____ adds _____. One response is _____.

Data source and date or capture time

Variables units and graph labels

A pattern supported by values

What the data cannot establish

Check the required evidence and revise one detail.

Stretch independent task

Complete this route only. Investigate causes and effects of climate change.

1. Describe a pattern in the supplied climate data.
2. Explain a cause and effect using the source.
3. Compare one response that tackles a cause with one that reduces harm.
4. Explain why a short data selection cannot establish every local effect.

Support: Choose the evidence independently. Use the frame only if it helps.

Data source and date or capture time

Variables units and graph labels

A pattern supported by values

What the data cannot establish

Explain why a short data selection cannot establish every local effect.

Exit ticket

Choose a response mode. You may give your answers privately to an adult.

What happens when fossil fuels are burned?

My response

How do emissions reduction and shade differ?

My response

Support: From ____ to ____, the data _____. Burning ____ adds _____. One response is _____.

Further challenge: Explain why a short data selection cannot establish every local effect.

One thing I changed or want to check
